



NLP Project part 1 (Progress summary)

Group 7

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Selected Baseline Models.

We select two open-source LLMs within the 0.3B–4B parameter range:

- 1- **Qwen 2.5** with (1.5B) parameter.
- 2- **Phi-3.5-mini** with (3.8B) parameter.

By using `baseline.py` we test and compare the two-baseline open-source LLMs on a sample medical question to verify their work before fine-tuning.

baseline.py job:

- 1- **Baseline verification** - Ensuring both candidate models work before selecting one for fine-tuning
- 2- **Quick comparison** - this to manually evaluate Turkish language support and decide on Qwen 2.5
- 3- **Memory testing** - Confirmed that both models can run sequentially on available hardware

Also, we create `batch_inference.py` to Automatically generates answers for multiple test questions and saves the results for comparison before vs. after fine-tuning.

batch_inference.py job:

- 1- **Template for batch evaluation**, Structure is ready for delivers the real 15,000+ question dataset.
- 2- **Baseline generation**, Will be used to:
 - Run all 1,500 test questions before fine-tuning.
 - Save results in CSV file format.
 - Provide baseline for comparison after fine-tuning.
- 3- **Evaluation tool**, we used this script to:
 - Re-evaluate the fine-tuned model on the same 1,500 questions.
 - Compare before vs. after results using ROUGE metrics.

Test Corpus Used.

While we are group number 7, odd number we used (alibayram/doktorsitesi) dataset.

From this dataset, a test set of 1,500 instances was created (matching the minimum project requirement). For baseline and fine-tuned evaluation, **100 samples** from this test set were used for detailed ROUGE-based comparison and error analysis due to computational constraints and time limitations. The remaining 1,400 instances are available but were not used for the detailed metric reporting, but we will use them all in final submission.

SFT Corpus Used.

Be using (alibayram/doktorsitesi) dataset first we start **Filtering**: Only specified medical fields were retained (per project requirements). when we check **Training set size**: about 38958 instances. After that we select **Test set size**: Minimum 1,500 instances. Note (Class distribution: Kept consistent between train and test splits and Format: Converted to ChatML format before fine-tuning).

Summary Table of Baseline Results.

Fine-tuning Qwen 2.5 - 1.5B on 15,000 Turkish medical QA pairs improved ROUGE-2 by 118% and reduced average response length by 79%, producing more focused and relevant medical answers. We wrote all our Observations in two xlsx files format (**comparison_summary.xlsx** and **error_analysis_sample.xlsx**)

comparison_summary.xlsx:

This file contains the quantitative evaluation results comparing the baseline model vs. the fine-tuned model. What It Measures, Performance metrics (ROUGE scores and answer characteristics) across the same 100 test samples for both models.

A	B	C	D	E	F	G	H
model	num_examples	avg_answer_len_chars	empty_answers	rouge1	rouge2	rougeL	rougeLsum
base	100	514.96	0	0.094382481	0.01144371	0.066253037	0.074217604
fine_tuned	100	108.26	0	0.099729098	0.024892634	0.08262109	0.082597264

metric	Baseline	fine-Tuned	Improvement
ROUGE-1	0.0944	0.1	(+6.4%)
ROUGE-2	0.0114	0.025	(+127%)
ROUGE-L	0.0663	0.083	(+25.8%)
Avg Response Length	515 chars	108 chars	-79% (more concise)

comparison_summary.xlsx

error_analysis_sample.xlsx:

This file contains qualitative error analysis showing specific examples of model outputs, their errors, and human judgments. What It Captures, A detailed breakdown of **where each model succeeded or failed on individual test questions.**

Example ID	Error Type	Fine-Tuned Issue
29	unsafe_recommendation	Said "1 week" for post-surgery sports (reference said 1 month)
51	wrong_treatment	Jumped to surgery instead of conservative treatment first
82	wrong_medical_info	Claimed disease exists when maybe not
30	hallucination	Gave unsupported percentages (70%)

error_analysis_sample.xlsx

Model Selected for Fine-Tuning.

Our selection (Qwen 2.5 - 1.5B)

We selected because of these reasons:

- 1- Better Turkish language support (critical for the doktorsitesi medical dataset, which is in Turkish).
- 2- Fits within hardware constraints (Colab).
- 3- Faster and more efficient fine-tuning compared to Phi-3.5-mini (3.8B).

Fine-tuning method: we use LoRA/QLoRA (parameter-efficient fine-tuning).

Training hardware: use Colab.

Training outcome: Loss reduced from 2.72 $\rightarrow$ 1.75; training time reduced from ~18 hours to ~1.5 hours.